# 1.5 Multivariable Calculus: Problems

Solutions are in solutions.md. Most problems are computational. A few (7, 8, 20–23) ask for short proofs.

## Warm-up

**1.** Let $f(x, y, z) = x^2 y + e^{yz}$. Compute $\nabla f$, and find the directional derivative of $f$ at $(1, 1, 0)$ in the direction of $(1, 2, 2)$.

**2.** Let $F(r, \theta) = (r\cos\theta,\ r\sin\theta)$. Compute the Jacobian matrix $JF$ and $\det JF$. Where is $DF$ invertible?

**3.** Find the tangent plane to the surface $z = x^2 + y^2$ at $(1, 2, 5)$. Find a normal vector to the surface $x^2 + y^2 - z = 0$ at the same point and check that the two answers agree.

**4.** Let $z = f(x, y)$ with $x = r\cos\theta$ and $y = r\sin\theta$. Use the chain rule to express $z_r$ and $z_\theta$ in terms of $z_x$ and $z_y$, and show that

$$z_x^2 + z_y^2 = z_r^2 + \frac{1}{r^2} z_\theta^2.$$

**5.** Evaluate $\iint_R (x + y)\, dA$, where $R = [0, 1] \times [0, 2]$.

**6.** Evaluate $\displaystyle\int_0^1 \int_x^1 e^{y^2}\, dy\, dx$ by reversing the order of integration.

## Core

**7.** (a) Let $f(x, y) = \dfrac{xy}{x^2 + y^2}$ for $(x, y) \neq 0$ and $f(0, 0) = 0$. Show that $f_x(0, 0)$ and $f_y(0, 0)$ exist, but $f$ is not continuous at $0$.

(b) Let $g(x, y) = \dfrac{x^2 y}{x^2 + y^2}$ for $(x, y) \neq 0$ and $g(0, 0) = 0$. Show that $D_u g(0)$ exists for every unit vector $u$, but $g$ is not differentiable at $0$.

**8.** Let $f : \mathbb{R}^2 \to \mathbb{R}$ have partial derivatives $f_x, f_y$ that exist near $a$ and are continuous at $a$. Prove that $f$ is differentiable at $a$.

**9.** Find and classify all critical points of $f(x, y) = x^3 + y^3 - 3xy$.

**10.** Find the maximum and minimum of $f(x, y, z) = x + 2y + 2z$ on the sphere $x^2 + y^2 + z^2 = 9$, using Lagrange multipliers. Check your answer with Cauchy–Schwarz.

**11.** The cylinder $x^2 + y^2 = 1$ meets the plane $x + y + z = 1$ in an ellipse. Find the highest and lowest points on this ellipse (the extreme values of $z$), using Lagrange multipliers with two constraints.

**12.** Compute $\iint_{\mathbb{R}^2} e^{-(x^2+y^2)}\, dA$ in polar coordinates, and deduce $\int_{-\infty}^{\infty} e^{-x^2} dx$.

**13.** Find the volume of the "ice-cream cone" region that lies inside the sphere $x^2 + y^2 + z^2 = 4$ and above the cone $z = \sqrt{x^2 + y^2}$.

**14.** Let $D$ be the parallelogram bounded by the lines $x + y = 0$, $x + y = 1$, $x - y = 0$ and $x - y = 2$. Evaluate $\iint_D (x + y) e^{x-y}\, dA$.

**15.** (a) Show that $F = (y,\, x)$ is conservative, find a potential, and compute $\int_C F \cdot dr$ for any curve $C$ from $(0,0)$ to $(1,2)$.

(b) Let $G = \dfrac{(-y,\ x)}{x^2+y^2}$ on $\mathbb{R}^2 \setminus \{0\}$. Show that $Q_x = P_y$, but $\oint G \cdot dr = 2\pi$ around the unit circle (counter-clockwise). Explain why there is no contradiction.

**16.** Use Green's theorem to show that the area enclosed by the ellipse $x^2/a^2 + y^2/b^2 = 1$ is $\pi ab$.

**17.** Compute the outward flux of $F(x,y,z) = (x,y,z)$ through the sphere of radius $R$ centred at $0$, (a) directly and (b) using the divergence theorem.

**18.** Verify Stokes' theorem for $F = (-y,\, x,\, 0)$ on the upper hemisphere $S = \{x^2+y^2+z^2 = 1,\ z \geq 0\}$ with upward normal. Compute both sides.

**19.** Compute the outward flux of $F = (x^3,\, y^3,\, z^3)$ through the unit sphere.

## Challenge

**20.** (a) Prove that if $f$ is $C^2$ on an open set in $\mathbb{R}^2$, then $f_{xy} = f_{yx}$. (Hint: consider $\Delta(h) = f(a+h,b+h) - f(a+h,b) - f(a,b+h) + f(a,b)$ and apply the mean value theorem twice.)

(b) Let $f(x,y) = \dfrac{xy(x^2-y^2)}{x^2+y^2}$ for $(x,y) \neq 0$ and $f(0,0) = 0$. Compute $\frac{\partial}{\partial y}\frac{\partial f}{\partial x}(0,0)$ and $\frac{\partial}{\partial x}\frac{\partial f}{\partial y}(0,0)$, and show that they differ.

**21.** Show that $\det : M_n(\mathbb{R}) \to \mathbb{R}$ is differentiable at $I$ with $D\det(I)(H) = \operatorname{tr} H$. Deduce that for invertible $A$, $D\det(A)(H) = \det A \cdot \operatorname{tr}(A^{-1}H)$.

**22.** Let $A$ be a real symmetric $n \times n$ matrix and $q(x) = x^t A x$.

(a) Show that $\nabla q(x) = 2Ax$.
(b) Use Lagrange multipliers on the unit sphere to show that $A$ has a real eigenvalue, and that $\max_{|x|=1} q(x)$ is the largest eigenvalue of $A$.

**23.** Let $F(x) = x/|x|^3$ on $\mathbb{R}^3 \setminus \{0\}$.

(a) Show that $\operatorname{div} F = 0$.
(b) Show that the outward flux of $F$ through the sphere $|x| = R$ is $4\pi$, for every $R > 0$.
(c) Explain why (a) and (b) do not contradict the divergence theorem. Show that the flux through *any* closed surface $S$ that bounds a nice region containing $0$ in its interior is $4\pi$.